

433.HW3

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Problem 1

```
sum_geom_series = function(x, n) { # parameters for x value and n terms
  total = 0 # initialize the total as 0
  for (i in 0:n) { # for loop that loops until the final term has been reached
    total = total + x^i # update the total by raising x to each value of i
  }
  return (total) # return the final total
}
sum_geom_series(0.5, 5) # output should be 1.96875
```

```
## [1] 1.96875
```

Problem 2

```
pascal_next = function(prevrow) { # function w/ parameter for previous row
  next_row = rep(0,length(prevrow)+1) # initialize next row, increase length
  for (i in 1:length(prevrow)) { # cycle through indices of previous row
    next_row[i+1] = prevrow[i]+prevrow[i+1] # update values in next row by
    # combining adjacent values in previous rows (hence i and i+1 indices)
  }
  next_row[1] = 1 # set first value of the row to 1
  next_row[length(next_row)] = 1 # set last value of the row to 1
  # note: this could be easily adjusted to a recursive function with parameter n
  # that generates n rows of Pascal's triangle
  return(next_row) # return the next row of the triangle
}
row = c(1,4,6,4,1) # define the previous row
pascal_next(row) # call function using previous row
```

```
## [1] 1 5 10 10 5 1
```

Problem 3

```

one_norm = function(matrix, rows, cols) { # parameters for data & # rows/columns
  mags = rep(0,cols) # create empty vector to contain magnitudes of each column
  for (i in 1:cols) { # for loop cycles through each column in the matrix
    mags[i] = sum(abs(matrix[,i])) # determine magnitude of each column by using
    # the built-in sum and absolute value functions, and update magnitude vector
  }
  return (max(mags)) # print out the maximum value in the magnitude vector
}
nrows = 3 # number of rows for matrix
ncols = 3 # number of columns for matrix
A = matrix(c(1,2,-5,-2,7,0,-10,3,-2),nrows,ncols, byrow = 1) # test matrix
one_norm(A, nrows, ncols) # call function with parameters

```

```
## [1] 13
```